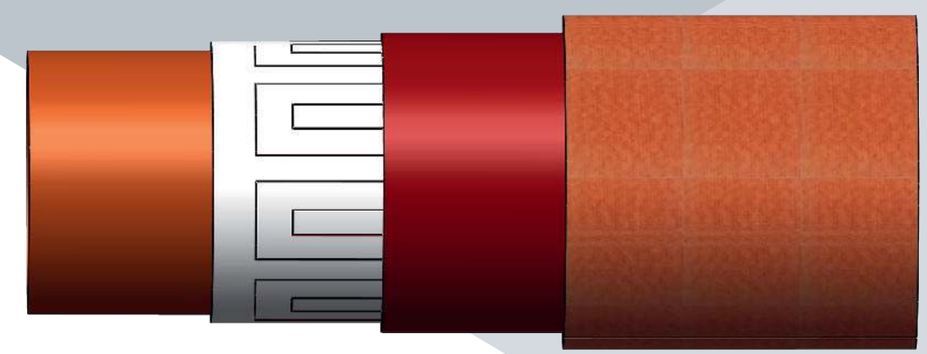
ANY QUESTIONS?
PLEASE FEEL FREE
TO CONTACT US!

SILICONE HEATING ELEMENTS

winkler



Robust flexible heating elements
Rail application / LSLT
UL / VDE / CSA

Winkler AG
Englerstrasse 24
69126 Heidelberg
Germany
Phone: +49 6221 3646 0
sales@winkler.org
www.winkler.org

© by WINKLER AG
2025-11 Version 1
All rights reserved.
Reproduction only with permission.

Delivery as long as in storage. Illustrations may be sample images that may differ in appearance from the delivered goods. Errors excepted. For misprints we accept no liability.

In this publication we avoid gender language for reasons of reading fluency and comprehension. In cases of gender-related salutation, all attitudes can feel addressed.

PRODUCT OVERVIEW SILICONE HEATING ELEMENTS

winkler

Select design

Flat

	❶ Flat	❷ Flat on assembly	❸ Flexible
Flexibility	○	○	Ø > 100 mm
Min. batch size	●	● ● ●	●
Protection class			

Cylindrical

	❹ Cylindrical and flexible	❺ Cylindrical	❻ spiralised
Flexibility	Ø > 20 mm...100 mm	Ø > 20 mm...100 mm	Ø 5 mm...50 mm
Min. batch size	● ● ●	● ● ●	● ● ●
Protection class			

Free form

	❼ Free form	❽ Free form arround assembly
Flexibility	○	○
Min. batch size	● ● ● ●	● ● ● ●
Protection class		

Select material

	Standard material	Extreme material
Models	CE LSLT UL94-V1	UL 94-HB/CE
Temperature range	-60° C...180° C	-60° C...200° C
Robustness / tensile strength	● ● ●	● ● ● ● ●
Advantages	■ Smooth surface	■ Structured surface ■ Particularly suitable for flexible designs

Select mounting

	RTV adhesive	Silicone band	Self-adhesive film	Tension springs	Eyelets/hooks	Silicone straps	Velcro fastener	Snap fasteners
Suitable designs	❶❷❸❹❺❻❼❽	❶❷❸❹❺❻❼❽	❶❷❸❹❺❻❼❽	❶❷❸❹❺❻❼❽	❶❷❸❹❺❻❼❽	❶❷❸❹❺❻❼❽	❶❷❸❹❺❻❼❽	❶❷❸❹❺❻❼❽
Heat transfer	● ● ● ● ●	● ● ● ●	● ● ● ●	● ● ●	● ● ● ●	● ● ●	● ●	● ●
Removability	○	○	○	● ●	● ● ●	● ● ● ●	● ● ● ●	● ● ● ● ●
Advantages	■ Universally applicable	■ Self-welding	■ Uniform adhesion	■ High clamping force ■ Self-tensioning	■ Robust and suitable for complex geometries	■ Self-tensioning	■ Cost-effective and practical	■ Easy assembly and disassembly

Select options

Control

No regulation

- self-stabilising
- Temperature controlled by heating output and ambient
- Temperature fluctuations possible

Temperature switch

- Breaks the circuit at the target temperature and closes it after cooling

Temperature controller

- Precise temperature control
- Temperature measurement using temperature sensors

Temperature sensor

- Accurate temperature measurement
- Thermocouple type K or type J
- Temperature sensor PT100 or PT1000

Sensor bag

- Temperature sensors can be inserted into sensor pockets
- Allows for simple exchange
- Alternatively, temperature sensors can be glued in

Security

Temperature switch

- Automatic shutdown at target temperature
- Multiple use

- Flat design of the temperature switch

- Temperature switch for lower temperatures
- More precise switching
- Only on request

- Durable design with a high number of switching cycles
- Only on request

Fuse

- Alternative to the temperature switch
- Protection against overheating
- Triggered when the temperature is too high
- One-way fuse

Plug

Stripped with wire end ferrules

- Easy wiring
- For applications without connectors

Round plug

- Plug and play connector with our temperature controllers
- Available in 3-, 4-, or 6-pin versions

AMP connector

- Connectors for applications with few cables

Molex connector

- Standard connectors for various applications

Harting connector

- For high-performance heating elements
- Wiring of multiple cables possible

Connections

Patch

- Flat connector
- Adjustable length and cross-section

T-Block

- Reduces stress on the connector
- Available in IP64 version

Head with strain relief

- Installation of temperature switches and fuses possible

Solder pad

- Connection option for individual adjustments

Thermal insulation

Standard insulation foam

- Talc-coated surface
- It is glued to the surface

UL insulation foam

- Textured surface
- UL certified
- Adhesively bonded to the surface

Pipe insulation

- Quickly snaps into place for easy installation
- Improves the efficiency of coil heaters
- Holds coil heaters in place without any fastening

General technical data

Dimensioning	max.B960 mm x L2800 mm
Thickness	min. 1,5 mm
Weight	min. 2 kg/m²
Operating voltage	12–480 V
Specific performance	max. 0,5 W/cm²
Degree of protection	max. IP65